SH Travel Hackathon Overview (Open)

Welcome to the open hackathon! This page contains everything you need to know about the project, along with links to essential resources

✍️ Project Description

The goal of our first hackathon is to create a fun and engaging app for travelling.

🎯 Goals

1. **Collaboration**: Take the opportunity to practice teamwork and effective communication when working with others.
2. **Skill Development**: Apply and sharpen your programming skills.
3. **Innovation**: Leverage creativity to design unique features.
4. **Enjoyment**: Have fun while building!

📋 Requirements

- 📚 **Travel-focused data and features**: Build an app that helps people plan or explore travel. Use suitable Python libraries to find destinations, calculate distances, display maps, explore routes, or plan itineraries.

Beginner-friendly Python libraries

Library	Difficulty	What students can build
`geopy`	🟢 Easy	Find city coordinates and calculate distances between destinations
`folium`	🟢 Easy	Create interactive travel maps with markers
`airportsdata`	🟢 Easy	Build an airport lookup or flight-distance calculator
`osmnx`	🟡 Intermediate	Find walking routes and explore city street networks
`ortools`	🟡 Intermediate	Plan an efficient multi-city itinerary

🧭 Getting Started

Start with the walkthrough example to get a feel for the app's flow. Then sketch the steps for your own idea and break your code into **small functions, each with one clear purpose**—for example, getting user input, finding destination coordinates, calculating distances, or displaying a map. Build and test one function at a time before connecting everything together.

🤝 Collaboration

You can participate **solo, in a duo, or as part of a team**. All three options are welcome! We encourage working with others, as the hackathon is a rare opportunity to collaborate, learn from one another, and build something together.

We strongly recommend using **Git** from the start of the hackathon. Create a shared repository on GitHub and commit your changes regularly. Git makes it easy for everyone to work on the project together, keeps track of changes, and provides a safety net if something goes wrong.

To help your team stay organized, consider using a simple project management tool such as **Trello**, **GitHub Projects**, or even a basic **Kanban board**. Breaking your project into small tasks and keeping them visible helps everyone stay on the same page, share the workload, and track your progress throughout the hackathon.

A few simple habits can make a big difference:

- Keep your task board up to date so everyone knows what's being worked on.
- **Always** `git pull` **before committing new changes** to make sure you're working with the latest version of the project.
- Commit small, meaningful changes often.

🤖 AI Policy: Consultant, Not Coder

While AI tools can be helpful, we recommend minimizing their use at this stage of your studies. They may replace the problem-solving process instead of supporting it, hindering your growth as a developer. A key part of the hackathon experience is tackling challenges, developing critical thinking skills, and collaborating as a team. This hands-on approach is essential for your development. Although AI can provide quick answers, relying on it may detract

from valuable learning opportunities. Embracing these challenges will deepen your understanding of the development process and enhance your skills.

Clarification on OpenAI Keys:
We can provide you with OpenAI API keys. These are intended to be used **inside your application logic** (e.g., if you want your app to generate summaries or answer user questions).

Using AI to Write Code:
⚠️ Using AI tools (like ChatGPT, GitHub Copilot, etc.) to **write the code for you is not allowed.** ⚠️

To grow as a developer, you need to build your own "coding muscle." While AI can provide a quick fix, it bypasses the critical thinking and debugging process that is an essential part of the hackathon.

The "Consultant" Rule:
If you get stuck, treat AI as a **consultant**, not a coder.

- ❌ **Don't ask:** "Write the code for my travel app."
- ✅ **Do ask:** "How should I break down the steps for calculating the distance between two destinations?" or "Can you explain this error I got while working with a mapping library?"

This is your chance to put your skills into practice and build the implementation yourself! Study Hall mentors are available for Q&A to help you work through questions along the way. If you are working in a duo or team, learn from one another and tackle challenges together!

🔑 Key resources

- **Travel app libraries:**
  - `airportsdata` — a simple starting point for airport lookups
  - `geopy` — find coordinates and calculate distances
  - `folium` — create interactive maps
  - `osmnx` — explore street networks and walking routes
  - `ortools` — plan multi-city itineraries
- **Walkthrough example - here**
- **Presentation example - here**

- **Workflow Chart Tools:**
  - draw.io
  - Canva
- **Presentation Templates:**
  - Final presentation template (copy from here)

For more information and other FAQs, please consult the Hackathon Intro Page